2B Terminating and Recurring Decimals

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. **Without dividing, state whether each simplified fraction converts to a terminating (T) or recurring (R) decimal:**

(a) $\frac{13}{40}$ (b) $\frac{7}{12}$ (c) $\frac{21}{56}$ (d) $\frac{11}{30}$

Q2. **Convert each fraction to an exact decimal (using dot or bar notation for recurring decimals):**

(a) $\frac{11}{16}$ (b) $-\frac{5}{6}$ (c) $\frac{7}{11}$ (d) $\frac{5}{7}$

Q3. **Convert each terminating decimal to a fraction in simplest form:**

(a) 0.45 (b) -0.375 (c) 1.64 (d) 0.0625

Q4. **Use algebra to convert each recurring decimal to a fraction in simplest form:**

(a) $0.\dot{7}$ (b) $0.\dot{4}\dot{5}$ (c) $0.\dot{1}3\dot{5}$ (d) $0.2\dot{7}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. In a Sheffield Shield cricket series, a bowler takes 15 wickets across 36 overs bowled. Express the wickets per over as (a) a simplified fraction and (b) a recurring decimal.

Q6. A laboratory balance in Parkville measures three chemical samples of masses $0.\dot{4}$ g, $0.\dot{2}\dot{7}$ g, and $0.\dot{1}$ g. By converting each to a fraction with denominator 99, find the exact total mass as a simplified fraction.

Q7. Arrange the following four rational numbers in ascending order (from smallest to largest): $0.6,\ 0.\dot{6},\ 0.\dot{6}\dot{0},\ 0.66$.

Q8. A steel bolt has a diameter of $\frac{11}{32}$ inches. Since $32 = 2^5$, explain why this fraction terminates and write its exact decimal value.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. A student claims that $\frac{18}{150}$ is a recurring decimal because 150 is divisible by 3. Explain the flaw in the student's reasoning and find the exact decimal.

Q10. Use the algebraic method ($x = 0.\dot{9},\ 10x = 9.\dot{9}$) to prove that $0.\dot{9} = 1$, and verify this result by adding $\frac{1}{3} = 0.\dot{3}$ and $\frac{2}{3} = 0.\dot{6}$.